

Some notes on the pool cooling model solution

Heat energy balance

$$\text{Rate energy accumulation} = - \text{Rate energy loss} \quad (w)$$

$$\dot{Q}_{sol} = q_{rad, sol} \cdot A_{top}$$

$$\dot{Q}_{top} = A_{top} \cdot U_{top} \cdot (T_{air} - T_{pool})$$

$$\dot{Q}_{sub} = A_{wall} \cdot U_{wall} \cdot (T_{sub} - T_{pool})$$

$$\dot{Q}_{renew} = c_p \cdot \rho \cdot \dot{V}_{renew} \cdot (T_{renew} - T_{pool})$$

$$\begin{aligned} \frac{dT}{dt} = & (q_{rad} \cdot A_{top} + A_{top} \cdot U_{top} \cdot (T_{air} - T_{pool}) \\ & + A_{wall} \cdot U_{wall} \cdot (T_{sub} - T_{pool}) \\ & + c_p \cdot \rho \cdot \dot{V}_{renew} \cdot (T_{renew} - T_{pool})) / (c_p \cdot \rho \cdot V) \end{aligned}$$

Isolate pool temperature, for both analytical dynamic model and steady state solution

$$\begin{aligned} \frac{dT}{dt} \cdot c_p \cdot \rho \cdot V = & q_{rad} \cdot A_{top} + A_{top} \cdot U_{top} \cdot T_{air} \\ & + A_{wall} \cdot U_{wall} \cdot T_{sub} \\ & + c_p \cdot \rho \cdot \dot{V}_{renew} \cdot T_{renew} \end{aligned}$$

$$(-A_{top} \cdot U_{top} - A_{wall} \cdot U_{wall} - c_p \cdot \rho \cdot \dot{V}_{renew}) T_{pool}$$

$$\frac{dT}{dt} = \left(-A_{top} \cdot u_{top} - A_{wall} \cdot u_{wall} - c_p \cdot \rho \cdot \dot{V}_{renew} \right) T_{pool} / c_p \cdot \rho \cdot V$$

$$+ \left(q_{rad} \cdot A_{top} + A_{top} \cdot u_{top} \cdot T_{air} + A_{wall} \cdot u_{wall} \cdot T_{sub} + c_p \cdot \rho \cdot \dot{V}_{renew} \cdot T_{renew} \right) / c_p \cdot \rho \cdot V$$

K

C

$$\frac{dT}{dt} = K \cdot T_{pool} + C$$

Steady-state solution

$$\frac{dT}{dt} = 0 \rightarrow -K \cdot T_{pool} = C$$

$$T_{pool} = -\frac{C}{K}$$

Parameter values

$$h = 100 \text{ W/m}^2 \cdot \text{K} \quad \text{or with no cover } U = 100 \text{ W/m}^2 \cdot \text{K}$$

$$q_{\text{sol}} = 0 \text{ at night or } \sim 200 \text{ W/m}^2 \text{ during spring or fall days}$$

$$U_{\text{wall}} = \frac{k}{L} = \frac{0.5 \text{ W/m} \cdot \text{K}}{0.15 \text{ m}} = 3.3 \text{ W/m}^2 \cdot \text{K}$$